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Hybrid joining of ultra-thin sheets via simultaneous laser shock clinching and welding

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Highlights

- A novel laser shock clinching and welding (LSCW) method for ultra-thin sheets
- Simultaneous clinching and welding are achieved using a single laser pulse
- LSCW joints replace the unbonded central region of LSW joints with clinching
- LSCW joints exhibit central clinching surrounded by annular welding
- LSCW joints exhibit improved mechanical performance by sequential load-bearing

Outline

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1. Introduction
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3. Numerical simulation
4. Experimental results
5. Evaluation of joint performance

Abstract

A novel hybrid joining method, laser shock clinching and welding (LSCW), is proposed for joining ultra-thin metal sheets. This integrates mechanical interlocking and solid-state metallurgical bonding in a single step by employing high-energy pulsed laser-induced shock waves. Experiments are conducted on Al/Cu and Cu/Cu combinations under varying laser energies and flying distances. The joints are characterized using optical microscopy, scanning electron microscopy, and energy-dispersive X-ray spectroscopy to examine the morphology and interfacial features. The LSCW joints exhibit a central circular clinched region surrounded by an annular welded region. A wavy interface and an elemental diffusion layer of about 5.42 μm are observed in the Al/Cu joint. Uniaxial shear tests and numerical simulations show that LSCW joints provide improved mechanical performance compared to those single-method joints, attributed to the sequential load-bearing of the clinched and welded regions. Specifically, the welding dominates the maximum load-bearing capacity, while the clinching enhances the total displacement. In addition, cross-sectional microhardness measurements show increased hardness at the welding interface. Electrical resistance tests indicate that the electrical conductivity of LSCW joints lies between those of single clinched and single welded joints, with a total resistance value of about 0.106Ω for Cu/Cu. These results confirm the feasibility and effectiveness of LSCW for microscale joining applications, particularly in microelectronics and lightweight components.

Graphical abstract

6. Conclusion

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgements

References

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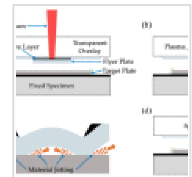
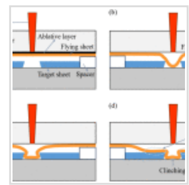
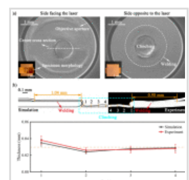
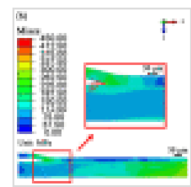
Figures (17)

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Tables (3)

Table 1. Laser indicators.

Table 2. The Johnson-Cook model param...

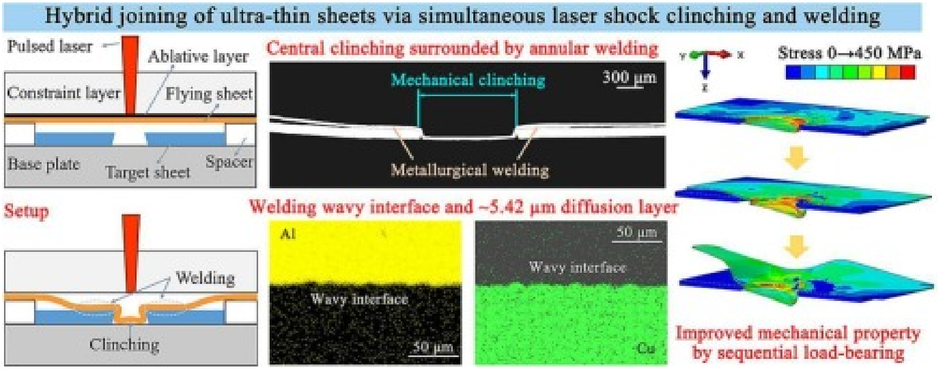

Table 3. Cohesive behavior and damage ...

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Keywords

Hybrid joining; Clinching; Welding; Ultra-thin sheets; Laser shock

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1. Introduction

The reliable joining of ultra-thin sheets is critical for the development of next-generation microelectronic devices, smart wearables, and medical devices. As products evolve towards miniaturization and functional integration, joining techniques are required to not only ensure structural integrity but also provide performance-related functionalities [1]. However, conventional joining techniques face inherent limitations when applied to ultra-thin sheets. Fusion welding such as laser welding, have demonstrated effectiveness in joining thin polymer sheets [2]. However, it is difficult to apply to thin metal sheets because of the heat affected zone and thermal damage [3,4]. Mechanical fastening such as self-pierce riveting may compromise fabrication accuracy [5]. Laser shock technology offers an alternative by using laser-induced shock waves as a driving force to achieve non-contact, high-precision, and high-strain-rate plastic deformation of ultra-thin sheets [[6], [7], [8]], which is considered green and sustainable [9]. Two joining techniques in this field are laser shock clinching (LSC) [10] and laser shock welding (LSW), also known as laser impact welding (LIW) [11].

In LSC, the upper sheet deforms plastically under high pressure and forms an interlocking structure constrained by a perforated lower sheet, thus realizing the mechanical joining [12]. Recent studies have expanded it to various configurations, including single-point joints [13], large-area linear joints [14], and adhesive-clinching hybrid joints [15]. However, LSC joints exhibit limited mechanical strength due to their reliance on geometric interlocking only, and poor electrical conductivity caused by the gap at the contact interface. In contrast, LSW employs high-speed impact welding to achieve solid-state metallurgical bonding [16]. It uses shock waves to drive intense plastic deformation of the flying sheet and impact on the substrate to form micro-solid-state welding [17]. Compared to conventional fusion welding, LSW avoids heat-affected zones and provides a closely bonded interface attributed to atomic forces. However, LSW joints exhibit poor energy absorption capacity and brittle failure under loading [18]. Moreover, interfacial debonding always occurs in the center region due to the propagation and superposition effects of stress waves, as shown in Fig. 1(d), welding occurs only in the annular region [16,19].

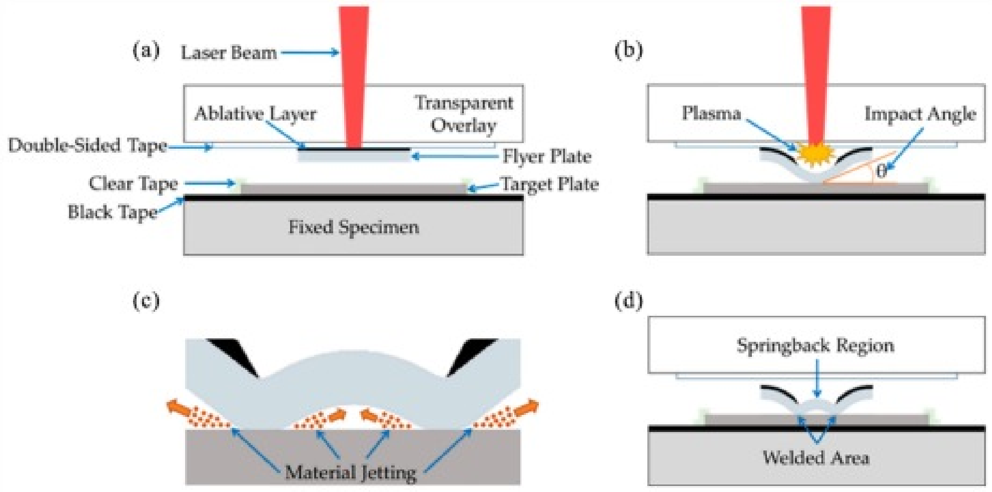

Fig. 1. Schematic diagram of LSW (a) Components (b) Laser shock (c) Material jetting (d) Springback region and welded area [16].

These complementary limitations of LSC and LSW highlight the absence of a technique that can simultaneously provide mechanical interlocking, metallurgical bonding, and excellent electrical conductivity for ultra-thin sheet joints. Zhang et al. [20] demonstrated the advantages of hybrid joining by combining riveting with resistance spot welding, but this technique, accomplished in two steps, retains all the limitations of each process and is limited to mm-thick plates. For ultra-thin sheets, replacing the springback region in the central region of LSW joints with the clinched region of LSC joints offers a viable solution. However, sequential hybrid strategies, including clinching before welding and welding before clinching, face process compatibility challenges. Pre-formed clinched structures can restrict welding flexibility and complicate the selection of welding positions and angles, while the material hardening caused by welding can increase the clinching difficulty and decrease the joint quality.

In this study, a laser shock clinching and welding (LSCW) hybrid joining technique is proposed. By spatially and temporally integrating clinching and welding with a single laser pulse, LSCW combines mechanical interlocking and metallurgical bonding. This paper is organized as follows: Section 2 describes the experimental setup, materials, and the deformation process of LSCW. Section 3 presents the establishment and validation of the numerical simulation. Section 4 discusses the experimental results, including process windows and joint characteristics. Section 5 evaluates the mechanical performance, microhardness, and electrical conductivity of the joints.

2. Experimental setup and materials

The device diagram of LSCW is shown in Fig. 2(a), which consists of a laser, a constraint layer, an ablative layer, a flying sheet, a target sheet, a spacer and a base plate. The spacer height determines the flying distance, a key parameter in joint formation. Simultaneous clinching and welding are achieved by a single pulse laser shock. Plasma shock waves generated by the interaction between the laser and the ablative layer propel the flying sheet downwards to undergo high-strain-rate plastic deformation [21]. The detailed deformation process is divided into three stages: (1) Initial bulging stage: The material in the center region of the flying sheet flows into the holes of the target sheet, forming a hemispherical bulge structure, as shown in Fig. 2(b); (2) Intermediate deformation stage: The flying sheet undergoes further horizontal deformation, forming the structure shown in Fig. 2(c). At this stage, the flying sheet forms a collision angle with the upper surface of the target sheet, providing the necessary conditions for impact welding; (3) Final stage. As shown in Fig. 2(d), the center region forms an interlocking structure to form a mechanically clinched region, while the outer regions create an annular metallurgical welding seam.

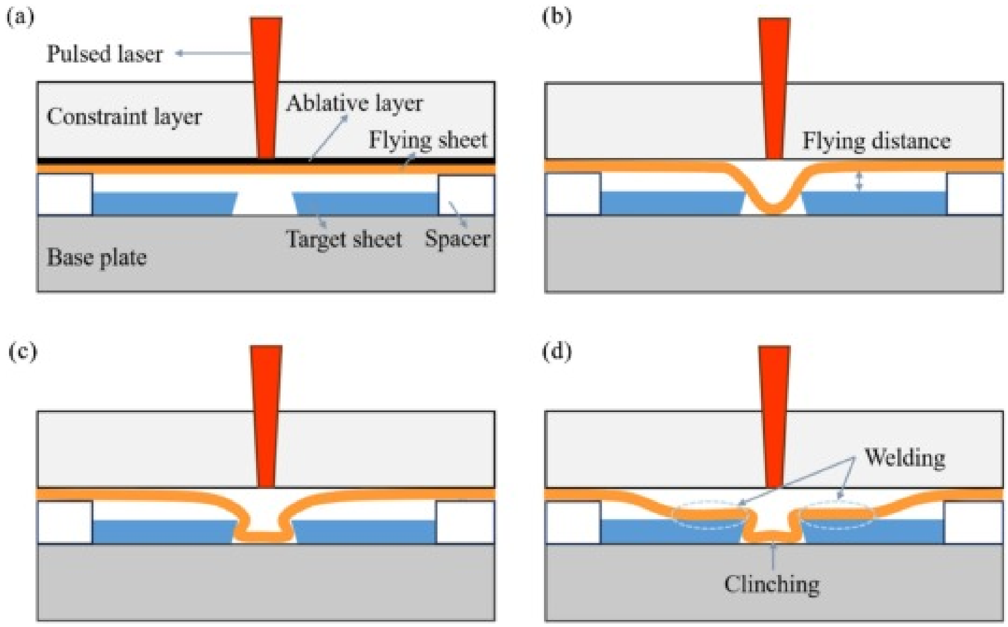

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Fig. 2. (a) Schematic diagram of LSCW (b) Initial bulging stage (c) Intermediate deformation stage (d) Final stage.

The SGR-60 Nd:YAG laser is employed to generate laser pulses in Q-switched mode. The laser beam passes through two reflective mirrors and a convex lens before focusing on the working platform with a preset spot diameter. The key laser parameters including output wavelength, output energy, pulse width, pulse frequency and spot diameter are listed in Table 1.

Table 1. Laser indicators.

Parameter	Output wavelength	Output energy	Pulse width	Pulse frequency	Spot diameter
Value	1064±1 nm	0.8–6 J	~12 ns	1 Hz	3.0 mm

A 3 mm thick K9 glass is used as the constraint layer to provide excellent light transmission and constraint effect, which can effectively constraint the propagation of shock waves and enhance the shock effect [22]. A black paint with a thickness of about 10 μm is uniformly sprayed on the side of the flying sheet facing the laser, to generate high-intensity shock waves and protect the material surface from ablation. The flying and target sheets are selected from 1060 pure aluminum and T2 pure copper foils, respectively. 1060 aluminum exhibits superior plastic deformability, while T2 copper offers higher strength and structural support. Both materials are widely used in microelectronic devices. Two material combinations are examined: similar Cu/Cu and dissimilar Al/Cu. In both cases, the flying sheet measures 25 mm × 10 mm × 0.03 mm, while the target sheet measures 20 mm × 20 mm × 0.1 mm and features a central tapered hole with diameters of 1.20 mm/1.28 mm. Both the spacer and the base plate are made of 304 stainless steels.

3. Numerical simulation

3.1. Simulation preparation

Since no significant temperature increase in LSCW is detected by the infrared handheld thermometer, a simplified Johnson-Cook model ignoring temperature changes is used [23]:

$$\sigma_\gamma = (A + B\varepsilon^n)\left(1 + Cln\,\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right) \tag{1}$$

where σ_γ denotes the equivalent von Mises stress, ε is the equivalent plastic strain, $\dot{\varepsilon}$ and $\dot{\varepsilon}_0$ are the current and reference strain rates, A, B, C and n are all material parameters. The specific parameters for copper foils are listed in Table 2.

Table 2. The Johnson-Cook model parameters of copper foil [24].

Parameter	A(MPa)	B(MPa)	C	n	$\dot{\varepsilon}_0$(s^{-1})
Value	89.63	291.64	0.025	0.31	1

The pressure model for laser shock waves refers to the one-dimensional formulation proposed by Fabbro et al. [25] and the peak pressure P is expressed as [26]:

$$P = 0.01\sqrt{\alpha/(2\alpha+3)\cdot Z\cdot I_0} \tag{2}$$

where Z is calculated from the acoustic impedances of the constraint layer Z_1 and the copper foil Z_2, as differences in the impedance of the constraint layer and the target material affect the reflection and transmission factors of the shock waves [27]. The specific expression is $2/Z = 1/Z_1 + 1/Z_2$. Z_1=1.38×10^7 kg/(m^2s), Z_2=3.42×10^7 kg/(m^2s), and Z=1.97×10^7 kg/(m^2s). α is the energy conversion factor and taken as 0.1 for glass confinement. I_0 is the laser power density, expressed as $I_0 = 4E/\pi d^2\tau$, where E is the laser energy, d is the laser spot diameter, and τ is the pulse width. The calculated I_0 are on the order of 10^{13}W/m^2 and the peak pressure can reach several GPa. The pressure pulse decreases more slowly than that of laser pulse due to the confinement effect of the plasma. Given that the LSCW process combines key characteristics of both LSC and LSW, adopting a triangular profile ensures consistency with previously established modeling approaches [28,29]. Moreover, the triangular pulse offers a physically meaningful and numerically efficient way to represent the short-duration and high-intensity pressure loading. By adjusting the pressure duration, the model can be calibrated to match the experimentally observed deformation features. As discussed in Section 3.2, the pressure duration in this simulation is set to 2.8τ based on experimental comparisons.

To enhance computational efficiency while maintaining accuracy, the simulation focuses only on the central 4mm×4mm region of the specimen. Fig. 3 shows a half-model of LSCW along the Y-direction to clearly present the geometry of the parts. The experimental constraint layer and base plate are simplified as rigid top and bottom dies in the model. Elastoplastic constitutive model and C3D8R elements are used for both the flying and target sheets. The central region of the flying sheet is set with a fine mesh of 0.015 mm and the rest is 0.04 mm. The mesh is refined to 0.04 mm at the inner wall of the hole and 0.08 mm at the periphery for the target sheet. All degrees of freedom for the dies are completely fixed to simulate the rigid-fixed state in the actual experiment. Face-to-face contact pairs are adopted to define the interaction between surfaces, with a friction coefficient of 0.25 [30], and hard contact is used for contact interaction property, allowing separation after contact.

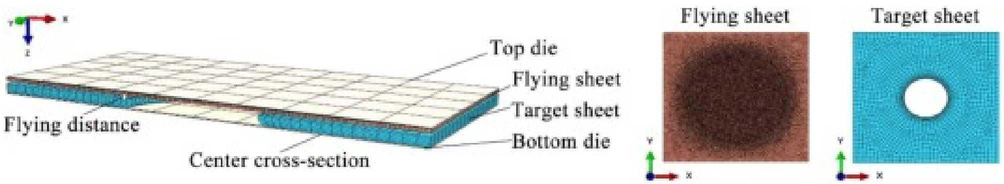

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Fig. 3. The LSCW finite element model and meshing.

The simulated forming results, such as deformed shape, stress and strain distributions, are imported into the initial state of the mechanical analysis by predefined fields for uniaxial shear simulation. A traction-separation law is used to describe the force-separation relationship at the interface to simulate the bonding behavior and the failure process [31]. In the linear elastic phase, the interfacial force state follows the linear relationship [32]:

$$t_{n,s,t} = K_{n,s,t} \cdot \delta_{n,s,t} \tag{3}$$

where $t_{n,s,t}$ and $\delta_{n,s,t}$ are stresses and displacement separations in the normal and tangential directions. $K_{n,s,t}$ denotes the interfacial stiffness in the corresponding directions. The quadratic stress criterion is adopted for the damage initiation criterion with the following expression [33]:

$$\left(\frac{\langle t_n \rangle}{t_n^{max}}\right)^2 + \left(\frac{t_s}{t_s^{max}}\right)^2 + \left(\frac{t_t}{t_t^{max}}\right)^2 = 1 \tag{4}$$

where $\langle t_n \rangle$ denotes that only normal stresses in tension are considered. During the damage evolution phase, the energy release rate G_c governs the failure process. The corresponding parameters are listed in Table 3.

Table 3. Cohesive behavior and damage model parameters.

K_n(MPa/mm)	K_s(MPa/mm)	K_t(MPa/mm)	t_n(MPa)	t_s(MPa)	t_t(MPa)	G_c(Nmm)
150	150	150	40	40	40	0.5

3.2. Simulation validation

Fig. 4(a) shows the macroscopic morphology and scanning electron microscopy (SEM) images of the experimentally fabricated joints of Cu/Cu at 0.2 mm flying distance and 2.0J laser energy. A circular objective aperture is visible in SEM images, this does not affect the observation of the joint morphology. The surface quality is excellent, with no evident signs of ablation, cracks or other damage. On the joint side directly facing the laser, a distinct feature of localized plastic flow is observed. A concave structure is formed in the central clinched region, and slight traces of plastic deformation are observed in the peripheral welded region. Notably, the transition between the clinched and welded regions is relatively smooth.

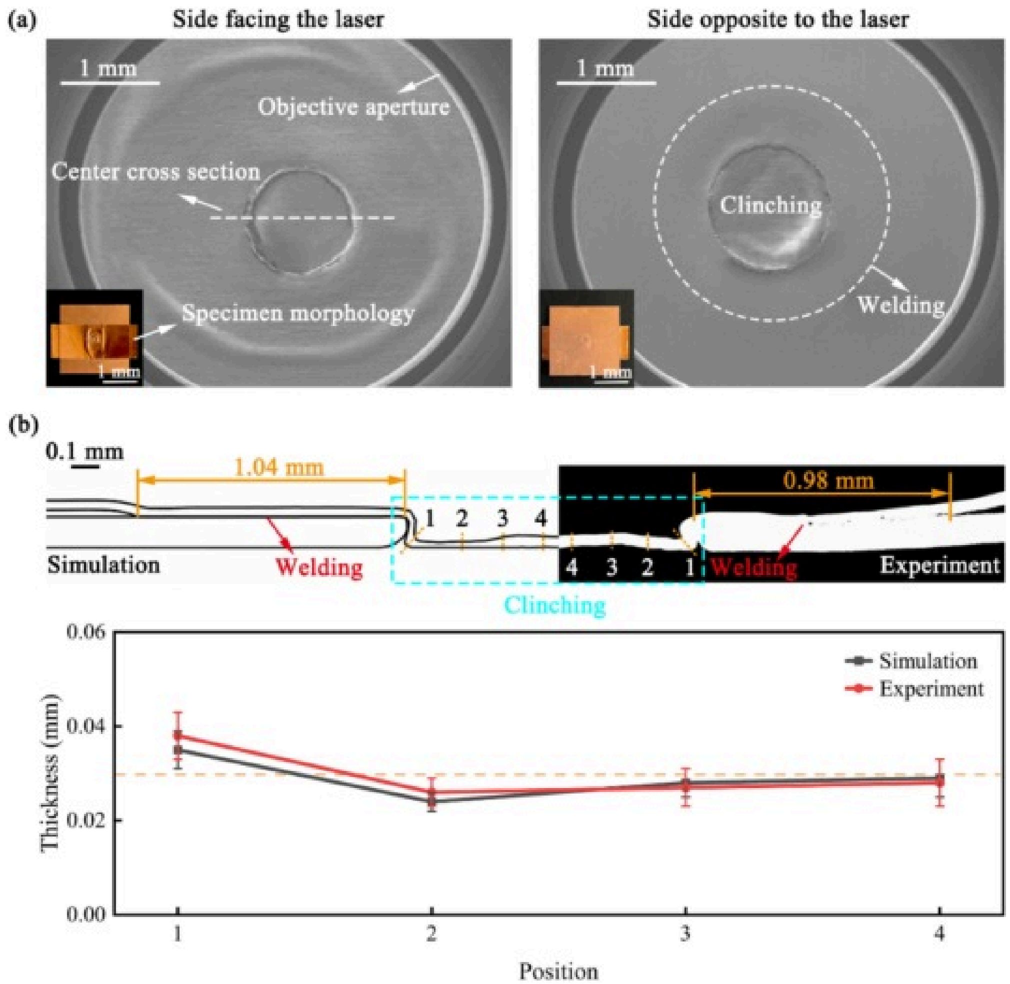

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Fig. 4. (a) Macro-morphology and SEM images of experimental LSCW joints (b) Comparison of experimental and simulated center cross-sections and thicknesses at four positions.

Fig. 4(b) presents the center cross-section indicated in Fig. 4(a) and compares the joint thicknesses at four positions. The experimentally measured length of the welded region is

0.98 mm, while the simulation shows 1.04 mm, a difference of 6.1%. Experimental data reveal a localized accumulation of material at Position 1, where the joint thickness reaches 0.038 mm. At Positions 2 and 3, material thinning occurs with thicknesses of approximately 0.026 mm and 0.027 mm. The joint thickness is nearly unchanged at Position 4. The general trend observed in the simulation is similar to the experiments, although the maximum difference is about 7.9% at position 1. This discrepancy may arise from measurement errors, simplifications in shock wave modeling and material constitutive modeling. Overall, the developed numerical model can predict the welding length and the clinching profile.

3.3. Welding validation

The fabrication of high-quality LSCW joints requires ensuring metallurgical effectiveness while creating interlocking structure. The impact velocity and collision angle of the flying sheet have been identified as critical parameters in solid-state impact welding [30]. The minimum impact velocity v_{min} can be estimated by the empirical formula [34]:

$$v_{min} = \sqrt{\frac{1.154\sigma_u}{\rho}} \tag{5}$$

where σ_u and ρ represent the ultimate tensile strength (230 MPa for copper foil in this study) and density of the flying sheet. The calculated threshold velocity is 172 m/s, compared to the simulated velocity of 210 m/s shown in Fig. 5(a), confirming the welding feasibility.

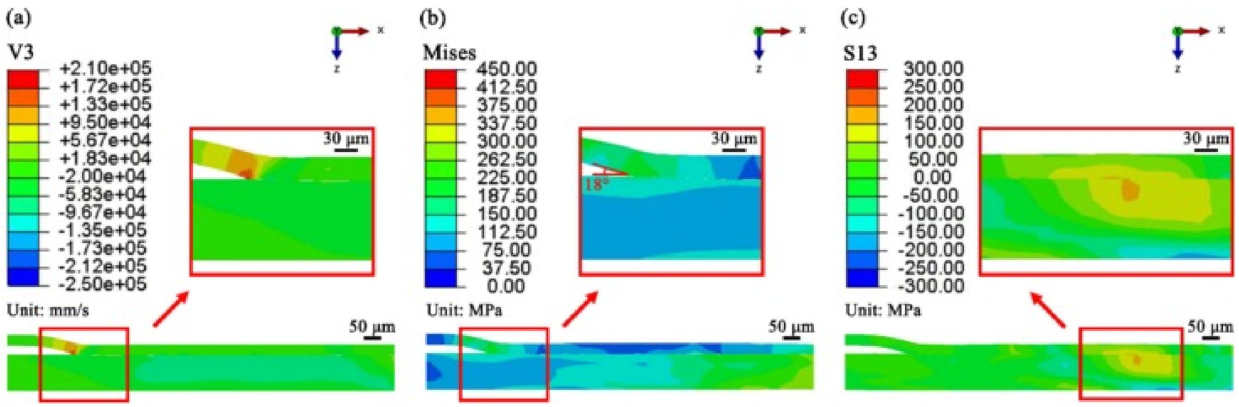

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Fig. 5. Simulated Cu/Cu LSCW joint cross-section (a) Velocity in the *Z*-direction (mm/s) (b) Mises stresses (MPa) (c) Shear stresses in the XZ-plane (MPa).

The minimum collision angle θ_{min} is inversely related to the impact velocity, expressed as [35]:

$$\theta_{min} = K\sqrt{\frac{H_v}{\rho \cdot v^2}} \tag{6}$$

where K is a material-dependent constant usually taken as 0.6–1.2, and H_v denotes Vickers hardness of the flying sheet. For the copper foil (K=1.0, H_v=90HV, ρ=8960kg/m^3, v=210m/s), the calculated θ_{min} equals 1.5°. The simulated collision angle in Fig. 5(b) is about 18°, which is larger than the threshold value and within the empirically validated reasonable range of 5–30°, confirming parameter applicability.

Further validation is performed according to Ref. [30], where interfacial shear stresses of the flying and target sheets exceed their shear yield stress values for effective welding. Applying the von Mises criterion ($\tau_{yield} = \sigma_u/\sqrt{3}$), the calculated shear yield stress for copper foil is 133 MPa. As shown in Fig. 5(c), the simulated interfacial shear stresses for both sheets exceed this threshold.

4. Experimental results

4.1. Process window

Fig. 6 presents the experimental results for Al/Cu and Cu/Cu material combinations at various flying distances and laser energies. The results are categorized into three types: success, failure, and damage. For the Al/Cu combination shown in Fig. 6(a), all flying distances exhibit material damage at 3.6 J energy, indicating high sensitivity to excessive energy input when using aluminum foils as flying sheets. After reducing the laser energy to 2.8 J, damage only occurs at a flying distance of 0.3 mm. When the laser energy is further decreased to 2.0 J, the 0.1 mm flying distance fails to achieve simultaneous clinching and welding, while successful joining is observed at flying distances of 0.2 mm and 0.3 mm. In contrast, Cu/Cu combination shown in Fig. 6(b) exhibits superior process adaptability. Even at a 3.6 J high energy, successful joining is achieved at a flying distance of 0.1 mm. However, when the flying distance is reduced to 0.1 mm, both laser energies of 2.0 J and 2.8 J show failure. This is attributed to the higher yield strength and strain hardening rate of copper foils compared to aluminum foils. Overall, both Al/Cu and Cu/Cu can achieve simultaneous clinching and welding with suitable parameters. Al/Cu has a narrower process window and is more prone to damage under high-energy input, while Cu/Cu offers better processing adaptability.

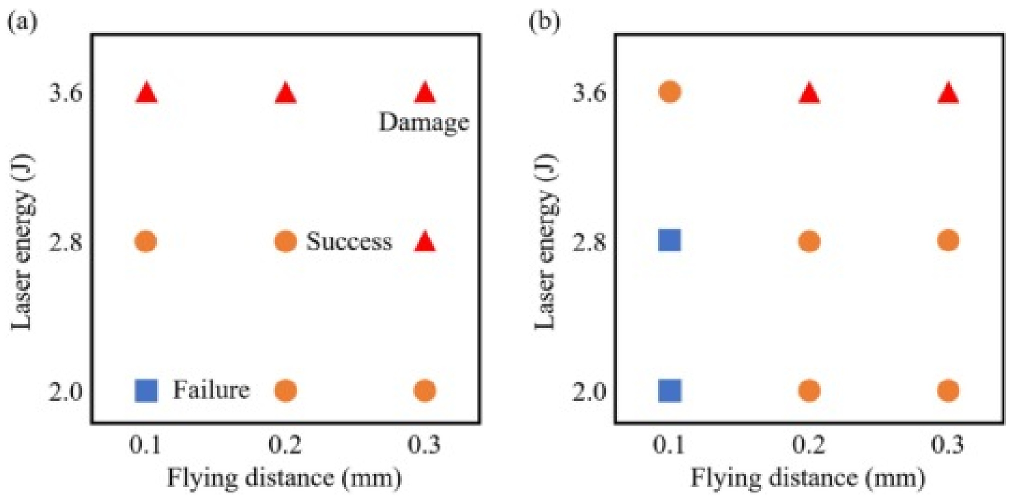

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Fig. 6. Experimental results at different flying distances and laser energies for (a) Al/Cu and (b) Cu/Cu.

4.2. Successful joining

Fig. 7 exhibits the surface morphology and center cross-section of a successful LSCW joint of Al/Cu formed at a flying distance of 0.1 mm and a laser energy of 2.8 J. The SEM images show a good surface quality of the hybrid joint, presenting a uniform metallic luster without visible ablation, cracks or other damage. A circular plastic deformation region, about 1.20 mm in diameter, is observed in the center region with smooth edge transitions. The cross-sectional view of the joint highlights a distinct and defect-free interface, confirming the structural integrity of the hybrid joint. It exhibits a hybrid joining characteristic of “central clinching surrounded by annular welding”.

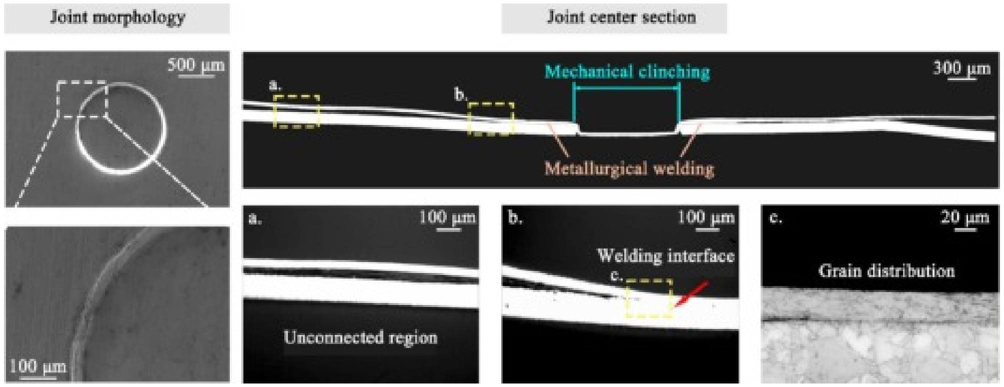

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Fig. 7. Morphology and center cross-section of a successful Al/Cu hybrid joint formed at 0.1 mm flying distance and 2.8J laser energy.

Specifically, in the central mechanical clinched region, the flying sheet flows into the preset hole under high pressure to form an interlocking structure. In the outer annular region, metallurgical welding occurs between the lower surface of the flying sheet and the upper surface of the target sheet, with an overall annular seam. Local magnified views provide further insight into the welding interface. In region a, located outside the laser shock affected region, an initial gap exists between the sheets. In contrast, region b shows that the laser shock force effectively eliminates the interfacial gap and achieves a close connection. In region c, the grain structure near the welding interface is displayed, showing significant grain refinement in the lower sheet adjacent to the interface. This microstructural feature is attributed to the intense plastic deformation and dynamic recrystallization induced by the laser shock [18].

The welding interface of Fig. 7 is further observed by SEM, as shown in Fig. 8(a). Distinct periodically undulated corrugations (commonly called wavy interface) are observed, with peak-to-valley spacing of 2.92–11.24 μm. This morphological feature is typical of high-speed shock welding processes and is attributed to the synergistic effects of ultra-high strain rates and ultra-high-pressure loading [[36], [37], [38]]. To examine the elemental diffusion behavior across the welded region, the energy-dispersive X-ray spectroscopy (EDS) line scanning is performed. The scanning path, indicated in Fig. 8(a), begins in the flying sheet and extends perpendicularly across the welding interface into the target sheet, with a total length of about 55 μm. As seen in Fig. 8(b), the EDS profiles exhibit smooth compositional transitions across the interface, characterized by an X-shaped cross. This suggests elemental diffusion between the two metals, and the thickness of the diffusion layer is 5.42 μm. In addition, no step-like or irregular features in the scanning curves indicate that no intermetallic compounds have formed at the interface. The formation of an elemental diffusion layer provides direct evidence of atomic migration, confirming metallurgical bonding.

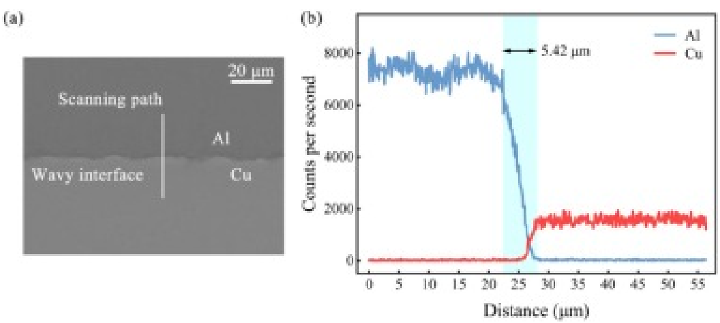

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Fig. 8. (a) Welding interface obtained by SEM (b) EDS line scanning analysis.

To further investigate the formation mechanism of the elemental diffusion layer, a simplified diffusion analysis is conducted based on the classical solution of Fick's second law [39]:

$$x \approx 2\sqrt{Dt} \tag{7}$$

where, x is the diffusion layer thickness, D is the solid-state diffusion coefficient, and t is the effective diffusion time. For the Cu—Al system, D typically ranges from $10^{-11}\,m^2/s$ to

$10^{-9} m^2/s$ under high temperatures [40]. According to Eq. (7), the minimum time required to form a 5.42 μm thick diffusion layer is about 7.34 ms, which is several orders of magnitude longer than the actual interaction time during laser shock loading (laser pulse width ~12 ns). Therefore, the formation of the elemental diffusion layer is not attributed to classical solid-state diffusion mechanisms. Instead, it is more likely the result of elemental mixing induced by material jetting or severe plastic deformation, similar to mechanical forced intermixing [18].

Fig. 9 shows the distribution of elements near the interface obtained by EDS surface scanning. The welding interface exhibits wavy characteristics. The Al foil (flying sheet) forms a wavy surface with an average curvature radius of 7.5±1.5 μm, while the Cu foil (target sheet) exhibits an average curvature radius of 9.5±2.0 μm. This results from the difference in mechanical properties and dynamic response of the two materials. Due to the lower yield strength, the Al foil experiences a jetting effect under high pressure, while the Cu foil's higher hardness and elastic modulus lead to localized material accumulation. Additionally, the acoustic impedance difference between the two materials also causes partial reflection of the shock wave at the interface. This reflection enhances the interfacial shear stress, promoting asymmetric plastic deformation and the formation of wavy characteristics. Moreover, a small amount of elemental diffusion is also observed away from the interface, caused by the injection of aluminum and copper jet particles during the high-speed collision. Overall, the synergistic mechanism of elemental diffusion and plastic deformation results in the welding interface of LSCW joints having both the mechanical interlocking structure and atomic-scale metallurgical bond, thus ensuring the comprehensive performance of welding.

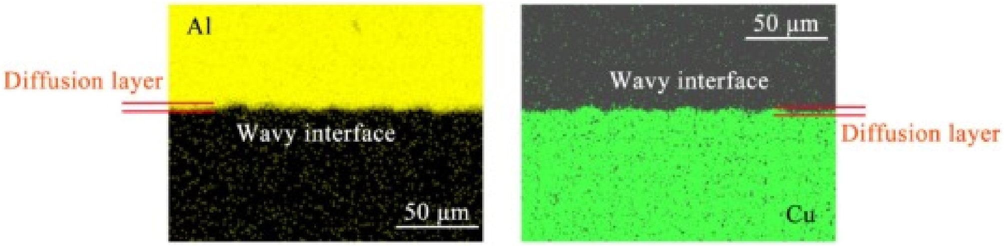

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Fig. 9. Element distribution at the welding interface obtained by EDS surface scanning.

4.3. Failed joining

The failure modes ("Failure" in Fig. 6) for LSCW joints include failed clinching, failed welding and both failures. Fig. 10 shows the surface morphology and center cross-section of failed joints. In Fig. 10(a), although partial welding is observed, the flying sheet fails to form an effective mechanical interlocking structure due to significant material rebound in the center region, resulting in failure of hybrid joining. Fig. 10(b) displays a case where both clinching and welding fail. Fig. 10(c) presents a shear fracture (belongs to "Damage" in Fig. 6). While the welded region is tightly bonded, the central material of the flying sheet experiences complete shear failure. This is attributed to excessive laser energy or flying distance, both of which intensify plastic deformation beyond the material's capacity, resulting in fracture.

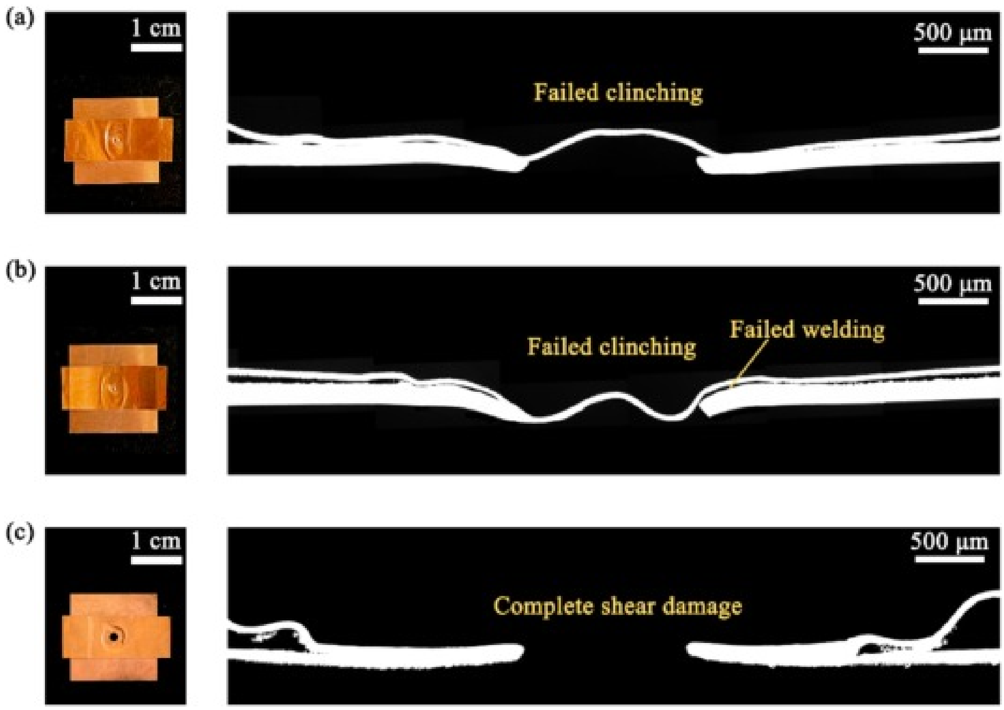

Fig. 10. Morphology and center cross-section of Cu/Cu failed hybrid joints at various process conditions (a) 0.1 mm flying distance and 2.8J laser energy (b) 0.1 mm flying distance and 2.0J laser energy (c) 0.3 mm flying distance and 3.6J laser energy.

5. Evaluation of joint performance

5.1. Mechanical property

Uniaxial shear tests are performed on LSC, LSW and LSCW joints of Cu/Cu to evaluate the shear resistance of joints with different techniques. The thickness combination between upper and lower sheets is 30μm/100μm, and the specimen dimensions are given in Fig. 11. The LSCW joints are produced at a flying distance of 0.2 mm and a laser energy of 2.8J.

Notably, LSW joints have the same flying distance and laser energy as LSCW joints, while the hole diameter of the perforated sheet and the laser energy of LSC joints are equal to those of LSCW joints. Shear tests are conducted on a stretching machine at a loading speed of 1 mm/s. To ensure the accuracy of the results, three specimens for each joint type are tested and the average values are used for analysis.

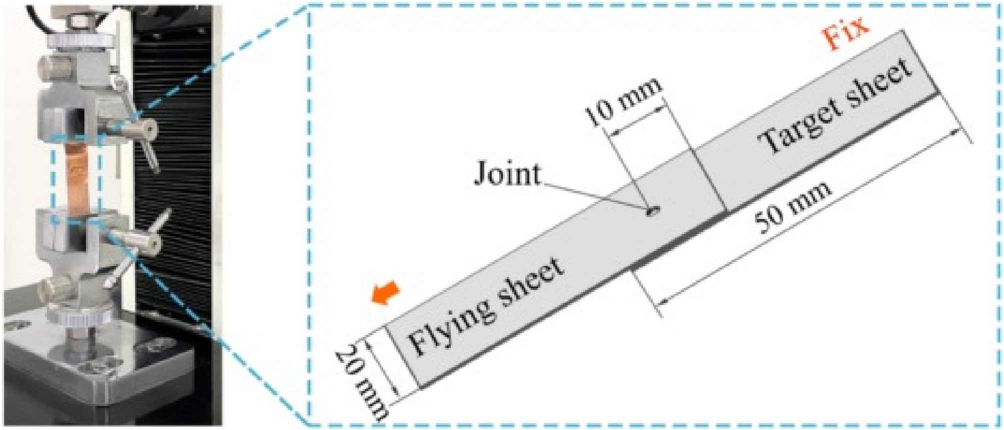

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Fig. 11. Joint uniaxial shear tests and specimen dimensions.

Fig. 12 presents the uniaxial shear test results for the three types of joints, where Fig. 12(a) shows the experimental and simulated force-displacement curves and the actual fracture morphology of the flying sheet in LSCW, and Fig. 12(b) compares the maximum force and absorbed energy (defined as the area under the force-displacement curve) in experiments. The good agreement between the experimental and simulated curves validates the reliability of the model, which is further used to investigate the detailed load-bearing process. As shown in the upper right corner of Fig. 12(a), typical ductile features are observed at the welding interface, including equiaxed and elongated dimples as well as fibrous tearing characteristics. These characteristics suggest a mixed fracture mode dominated by ductile tension-shear failure. Such behavior is consistent with that of

LSW joints under high energy inputs [18], where the failure mode exhibits ductile fracture within the base material.

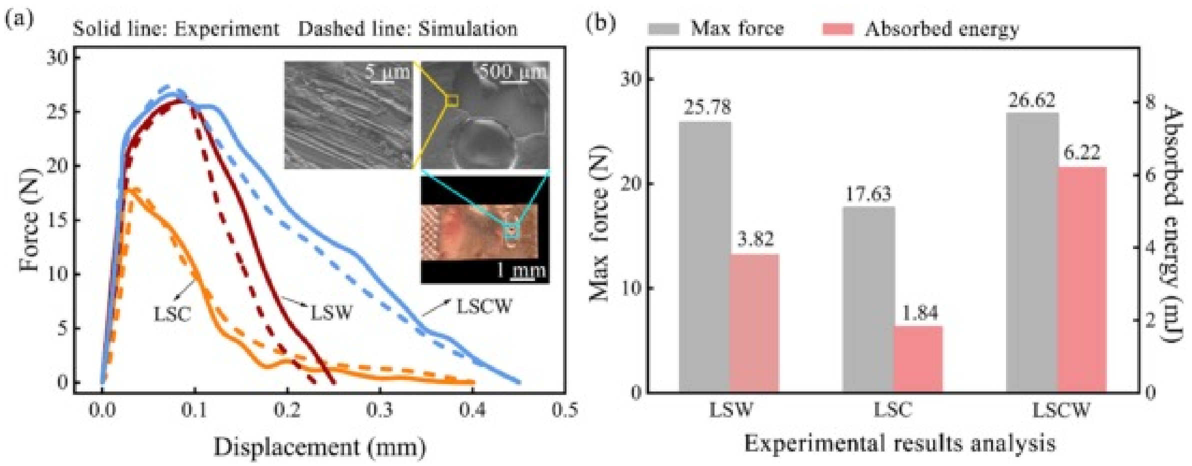

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Fig. 12. Uniaxial shear test results for three types of joints (a) Experimental and simulated force-displacement curves and actual fracture morphology of the flying sheet in LSCW (b) Comparison of maximum forces and absorbed energies in experiments.

For experiments, the LSW joint exhibits higher load-bearing capacity, and its force-displacement curve decreases at a faster rate after reaching a peak value of 25.78 N, with a total displacement of only 0.25 mm. This is attributed to the brittle fracture characteristics of the metallurgical bonding interface. Once the applied load exceeds the bonding strength, cracks quickly propagate along the interface, leading to abrupt joint failure. In contrast, the LSC joint reaches a lower peak load of 17.63 N but exhibits a greater total displacement of 0.40 mm. This is because stress concentrations in the neck of the clinched region result in an earlier peak load, while continued plastic deformation leads to a greater total displacement. The LSCW joint, integrating both clinching and welding, shows improved mechanical performance. It has a peak load of 26.62 N, a 3.26% increase over LSW and a 50.99% increase over LSC. Furthermore, the LSCW joint absorbs the highest energy at 6.22 mJ, which is 1.63 and 3.38 times that of LSW and LSC joints, respectively.

For simulations, Fig. 13 presents the Mises stress distributions during load-bearing process for the three joint types. Compared to joints with a single joining method, LSCW joints exhibit a unique multi-stage load-bearing pattern: (1) Stage 1, localized failure initiates at the welding interface when the displacement reaches 0.075 mm, while the clinched region remains plastically deformed and undamaged; (2) Stage 2, at a displacement of 0.13 mm, damage occurs in the neck clinched region in contact with the inner wall of the hole. Both the clinched and welded regions contribute to load-bearing; (3) Stage 3, further fracture occurs in the neck and welded regions, eventually leading to complete separation of the flying sheet from the target sheet.

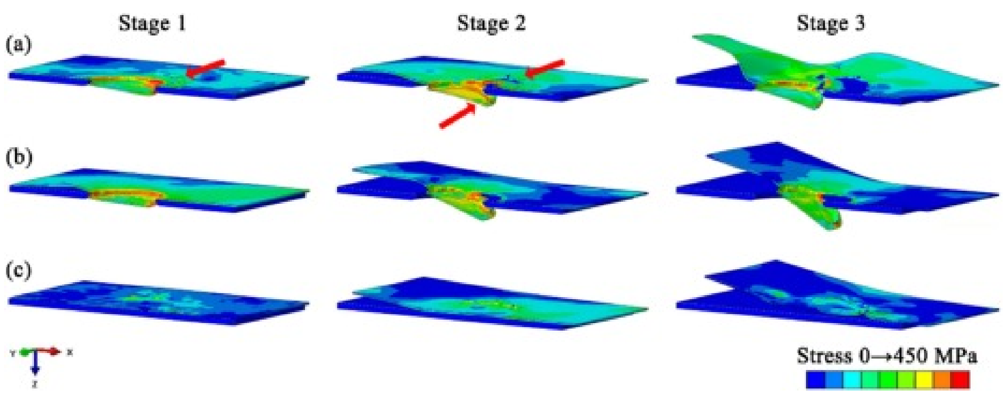

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Fig. 13. Load-bearing process for three types of joints (a) LSCW (b) LSC (c) LSW.

Combining the experimental and simulation results, Fig. 14 presents a schematic diagram of crack propagation during the load-bearing process for the three joint types, aimed at illustrating the improved mechanical performance of LSCW joints. For LSC joints shown in Fig. 14(b), cracks occur at the neck of the clinched region due to direct contact with the inner wall of the hole. This results in a relatively low load-bearing capacity, despite allowing for a larger total displacement. LSW joints shown in Fig. 14(c) exhibit higher load peaks, with cracks typically propagating along the welding interface. Compared to LSW joints, the clinching in LSCW joints shown in Fig. 14(a) participates in bearing part of the shear force, which delays crack propagation and contributes significantly to energy absorption. In addition, the welded region primarily dominates the maximum load-bearing capacity of the LSCW joint, while the clinched region contributes to the total displacement. The LSCW joint achieves co-strengthening by sequential load-bearing of the clinched and welded regions.

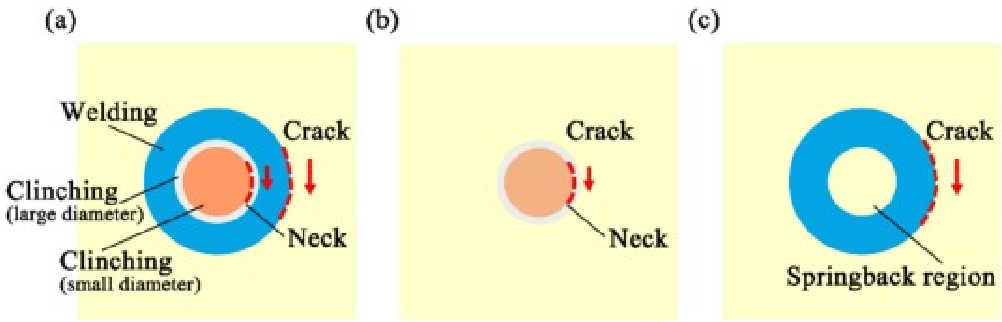

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Fig. 14. Schematic of crack propagation during load-bearing for three types of joints (a) LSCW (b) LSC (c) LSW.

5.2. Microhardness

The microhardness of LSCW joints is evaluated by a THV-1MDT microhardness tester. Measurements are carried out along the direction normal to the welding interface at the joint center cross-section, with the interface defined as the zero point, over a range of −20 μm to 40 μm. A test load of 100 gf (0.98 N) is applied with a hold time of 10 s. Each point is measured three times and averaged to minimize error.

For the Al/Cu LSCW joint shown in Fig. 15(a), the microhardness distribution exhibits an asymmetric gradient. The initial hardness in the Al foil region (from −20 μm to the interface) is about 52.9HV, which gradually increases to 77.6HV as approaching the interface. Near the interface (~0μm), the hardness reaches a peak of 90.9HV, and in the Cu foil region, the hardness decreases to about 88.2HV. The hardness distribution of the Cu/Cu LSCW joint is illustrated in Fig. 15(b). The hardness values at −20μm and 20μm from the interface are 90.3HV and 92.1HV, while those at −10μm and 10μm are 109.2HV and 116.4HV. This symmetric hardening pattern results from the similar acoustic impedance and material properties of the two Cu foils, allowing for bidirectional and nearly non-reflective shock wave transmission. The peak hardness value at the interface (~0μm) reaches 120.8HV. Notably, the hardness value at 40μm from the interface is 88.5HV, which closely matches the 88.2HV in the Cu foil region of the Al/Cu joint, indicating this is the original hardness of the Cu foil.

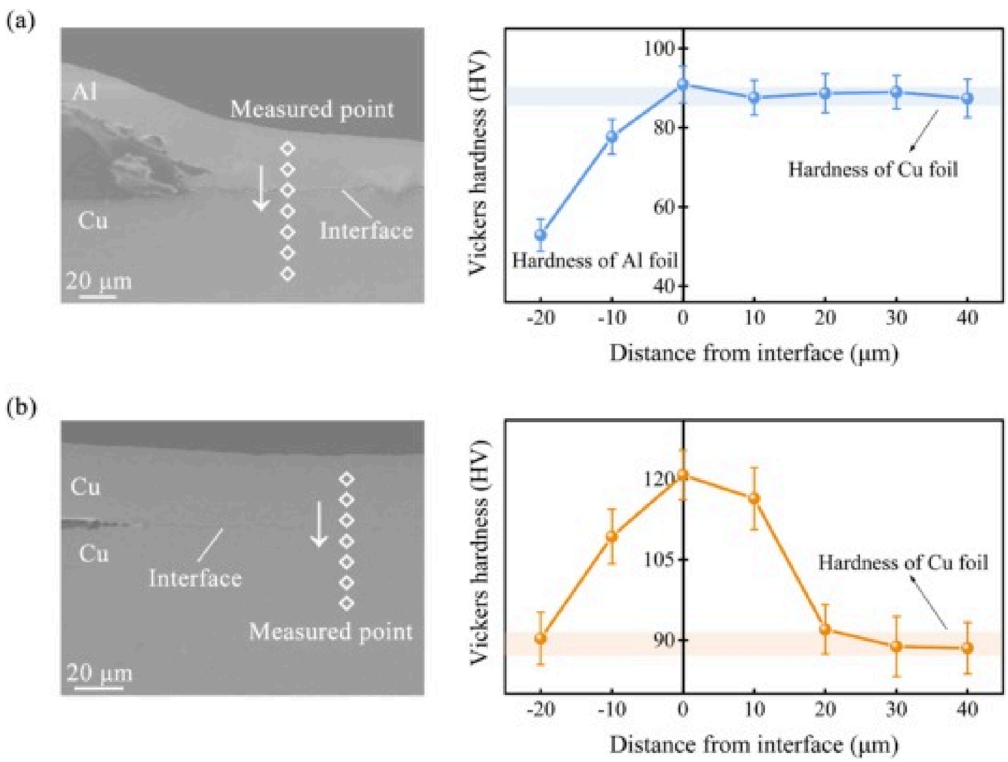

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Fig. 15. Microhardness measurement positions and results for LSCW joints (a) Al/Cu (b) Cu/Cu.

The above analysis shows that LSCW significantly enhances the microhardness near the interface. This can be attributed to multiple mechanisms associated with high-strain-rate plastic deformation. First, the severe plastic deformation induced by the laser shock leads to grain refinement, thus increasing the hardness according to the Hall-Petch relationship [41]. Second, the shock loading greatly increases the dislocation density, further leading to hardening effect. Additionally, although no significant temperature rise is detected by the infrared thermometer, localized adiabatic heating may also cause microstructural evolutions such as dynamic recovery and recrystallisation, leading to increased hardness.

5.3. Electrical conductivity

The electrical conductivity of LSC, LSW and LSCW joints of Cu/Cu is evaluated by measuring the electrical resistance. A four-wire resistance measurement method, also known as the Kelvin connection method, is used to eliminate the influence of wire and contact resistances. The target sheet (“Foil 1”) and the flying sheet (“Foil 2”) are 50 mm×20 mm copper foils with thicknesses of 100 μm and 30 μm, consistent with the dimensions of Fig. 11. The actual resistivities of the two are calculated to be $5.60\times10^{-7}\,\Omega\cdot\mathrm{m}$ and $5.52\times10^{-7}\,\Omega\cdot\mathrm{m}$, based on $R=\rho\cdot L/A$, where R is the resistance, ρ is the resistivity, L is the length, A is the cross-sectional area. The close match between the two foils confirms the feasibility of this measurement method. The difference between the actual and theoretical resistivities arises from surface oxides and internal impurities.

The specimens for all three types of joints, LSC, LSW, and LSCW, are made from the same batch of foils and are considered to have the same actual resistivity. For each type of joint, three specimens are prepared, each of which is measured three times independently and the average value is taken as the final result. The total resistance measured includes the bulk resistance of the foils and the contact resistance at the joint interface. As the bulk resistance is the same for all specimens, the measured results can be used to evaluate the joint contact resistance.

As shown in Fig. 16, the LSC joint exhibits the highest electrical resistance (0.168 Ω), the LSW joint shows the lowest (0.063 Ω), and the LSCW joint lies in between (0.106 Ω). The high resistance observed in the LSC joint is primarily due to its mechanical nature, where the not fully continuous physical contact leads to an increased contact resistance. In contrast, the LSW joint forms metallurgical bonding with atomic-level contact and produces a jetting effect to remove oxides and contaminants from the material surface, thereby reducing interfacial resistance.

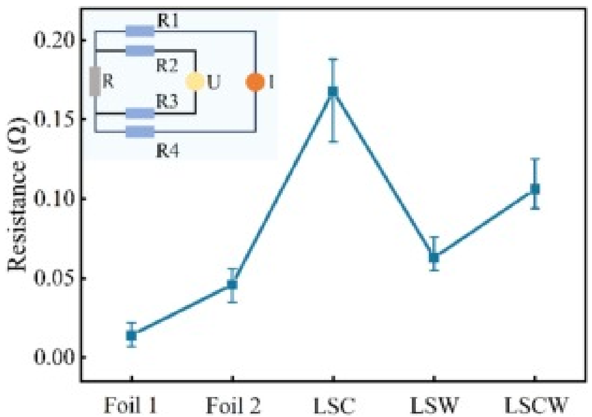

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Fig. 16. Total resistance values for the three joint types, including bulk and contact resistance.

The LSCW joint, which combines clinching and welding, shows an intermediate resistance. Notably, this value exceeds the sum of the bulk resistance (0.061 Ω) by approximately 73.8%. This is primarily attributed to interfacial effects. Specifically, the mechanical clinching may retain a physical gap or imperfect contact, leading to localized high contact resistance. Surface oxides and contaminants on the foils may further increase the interfacial resistance. Moreover, the elemental diffusion layer formed at the interface usually has a higher electrical resistivity than the base metal, further increasing the overall resistance. Several mitigation strategies can be considered to reduce the joint resistance. Applying surface pretreatment methods such as laser cleaning prior to joining can remove oxides and contaminants. Additionally, optimizing the laser shock parameters can improve jetting effects and the quality of metallurgical bonding at the interface.

6. Conclusion

In this study, a hybrid joining method for ultra-thin sheets called laser shock clinching and welding (LSCW) is proposed, which integrates clinching and welding spatially and temporally using a single pulse laser shock. Process windows for Al/Cu and Cu/Cu joints are established, with criteria defined for successful joints. The mechanical properties and load-bearing modes of LSCW joints are examined by combining experiments and simulations. The microhardness and electrical conductivity are also assessed. The main conclusions are as follows:

(1) LSCW joints can be successfully fabricated for both Al/Cu and Cu/Cu at suitable laser energies and flying distances, with the Cu/Cu showing superior process adaptability.

(2) Successful LSCW joints feature a central circular clinched region surrounded by an annular welded region. An elemental diffusion layer about 5.42 μm thick is observed at the wavy welding interface of the Al/Cu joint.

(3) Compared to LSC and LSW joints, LSCW joints provide improved mechanical performance in uniaxial shear tests through sequential load-bearing of the clinched and welded regions.

(4) The welded region primarily dominates the maximum load-bearing capacity of the LSCW joint, while the clinched region contributes to extending the total loading displacement.

(5) The microhardness at the welding interface is enhanced compared to the base material, with values of 90.9 HV and 120.8 HV for the Al/Cu and Cu/Cu joint interfaces.

(6) The total resistance of Cu/Cu LSCW joints (0.106 Ω) lies between that of LSC and LSW joints, and exceeds the bulk resistance (0.061 Ω) due to contact resistance and interfacial effects.

CRediT authorship contribution statement

Yaxuan Hou: Writing – review & editing, Writing – original draft, Investigation. **Yanran Wu:** Validation. **Zhong Ji:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] G. Meschut, M. Merklein, A. Brosius, *et al.*
Review on mechanical joining by plastic deformation
J Adv Joining Processes, 5 (2022), Article 100113
View PDF View article View in Scopus ↗ Google Scholar ↗

[2] D. Oguzhan, Ali A. Ahmed, M. Marco

Characterization of polypropylene pulsating heat stripes: effects of orientation, heat transfer fluid, and loop geometry
Appl Therm Eng, 184 (2021), Article 116304
Google Scholar ↗

[3] A. Sadeghian, N. Iqbal
A review on dissimilar laser welding of steel-copper, steel-aluminum, aluminum-copper, and steel-nickel for electric vehicle battery manufacturing
Opt Laser Technol, 146 (2022), Article 107595
View PDF View article View in Scopus ↗ Google Scholar ↗

[4] J. Yang, J.P. Oliveira, Y.L. Li, *et al.*
Laser techniques for dissimilar joining of aluminum alloys to steels: a critical review
J Mater Process Technol, 301 (2022), Article 117443
View PDF View article View in Scopus ↗ Google Scholar ↗

[5] J.L. Wu, C. Chen, Y.W. Ouyang, *et al.*
Recent development of the novel riveting processes
Int J Adv Manuf Technol, 117 (2021), pp. 19-47
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[6] H.F. Lu, L.J. Wu, H.L. Wei, *et al.*
Microstructural evolution and tensile property enhancement of remanufactured Ti6Al4V using hybrid manufacturing of laser directed energy deposition with laser shock peening
Addit Manuf, 55 (2022), Article 102877
View PDF View article View in Scopus ↗ Google Scholar ↗

[7] Z. Zhang, W.Z. Qiu, G.J. Zhang, *et al.*

Progress in applications of shockwave induced by short pulsed laser on surface processing
Opt Laser Technol, 157 (2023), Article 108760
View PDF View article View in Scopus ↗ Google Scholar ↗

[8] Y.Q. Yang, H.C. Qiao, Y. Lu, *et al.*
The plastic flow mechanism and precise forming method of 7075 aluminum plate in laser shock forming
Opt Laser Technol, 167 (2023), Article 109823
View PDF View article View in Scopus ↗ Google Scholar ↗

[9] Edward H.Z. Yan, F. Guo, B.L. Zhang, *et al.*
Sustainability assessment during machining processes: evidence from the econ-environmental modelling
J Clean Prod, 448 (2024), Article 141612
View in Scopus ↗ Google Scholar ↗

[10] Y.X. Hou, Z. Ji, H.M. Zhang
Crystal plasticity modeling for grain size and texture effects on laser shock clinching of ultra-thin foils
J Mater Res Technol, 35 (2025), pp. 3293-3303
View PDF View article View in Scopus ↗ Google Scholar ↗

[11] G.S. Daehn, J.C. Lippold
Low-temperature spot impact welding driven without contact
The Ohio State University (2009)
WO2009036499
Google Scholar ↗

[12] X.Y. Wang, Z. Ji, J.F. Wang, *et al.*

An experimental and numerical study on laser shock clinching for joining copper foil and perforated stainless steel sheet
J Mater Process Technol, 258 (2018), pp. 155-164
View PDF View article Google Scholar ↗

[13] X.Y. Wang, Z. Ji, C. Zheng, *et al.*
Joining similar and dissimilar material combinations by laser shock forming
J Manuf Process, 60 (2020), pp. 318-327
View PDF View article View in Scopus ↗ Google Scholar ↗

[14] J. Wang, Y.Q. Wang, S. Wang, *et al.*
Experimental and numerical investigation on incremental laser shock clinching for joining three sheets of copper/aluminum/stainless steel
Opt Laser Technol, 141 (2021), Article 107141
View PDF View article View in Scopus ↗ Google Scholar ↗

[15] Y.Q. Wang, G.X. Lu, Z. Ji, *et al.*
The small-size laser shock adhesive-clinching of Al foils
Int J Adv Manuf Technol, 122 (2022), pp. 3837-3848
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[16] S. Sadeh, G.H. Gleason, M.I. Hatamleh, *et al.*
Simulation and experimental comparison of laser impact welding with a plasma pressure model
Metals, 9 (2019), Article 1196
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[17] S. Sadeh, A. Malik
Investigation into the effects of laser shock peening as a post treatment to laser impact welding
Mater Des, 205 (2021), Article 109701

View PDF View article View in Scopus ↗ Google Scholar ↗

[18] K.N. Wang, H.M. Wang, H.Y. Zhou, *et al.*
Research status and prospect of laser impact welding
Metals, 10 (11) (2020), Article 1444
Crossref ↗ Google Scholar ↗

[19] Q.H. Wang, K.N. Wang, H.M. Wang, *et al.*
Variety of microstructure and mechanical properties along the laser impact welded interface
Sci Technol Weld Join, 28 (2023), pp. 128-136
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[20] Y. Zhang, C.M. Wang, H. Shan, *et al.*
High-toughness joining of aluminum alloy 5754 and DQSK steel using hybrid clinching-welding process
J Mater Process Technol, 259 (2018), pp. 33-44
View PDF View article Crossref ↗ View in Scopus ↗ Google Scholar ↗

[21] H.P. Yan, Z.Y. Qin, W. Zhang, *et al.*
Dynamic analysis of laser shock response: experimental and numerical studies
Aerosp Sci Technol, 94 (2019), Article 105430
View PDF View article View in Scopus ↗ Google Scholar ↗

[22] T. Shen, X.X. Zhao, J. Li, *et al.*
Laser shock incremental forming by employing high repetition rate laser pulses with pulse duration in the scale of 100 ns
Int J Adv Manuf Technol, 133 (2024), pp. 1393-1403
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[23] X.Y. Liu, Z.Y. He, J.Y. Ye, *et al.*

Study on dynamic mechanical behavior of Q460JSC and HQ600 high strength steel
J Constr Steel Res, 173 (2020), Article 106232
View PDF View article View in Scopus ↗ Google Scholar ↗

[24] X. Wang, H.F. Zhang, Z.B. Shen, *et al.*
Experimental and numerical investigation of laser shock synchronous welding and forming of copper/aluminum
Opt Lasers Eng, 86 (2016), pp. 291-302
View PDF View article View in Scopus ↗ Google Scholar ↗

[25] R. Fabbro, J. Fournier, P. Ballard, *et al.*
Physical study of laser-produced plasma in confined geometry
J Appl Phys, 68 (1990), pp. 775-784
View in Scopus ↗ Google Scholar ↗

[26] W.N. Hu, X.Y. Peng, Y. Ding, *et al.*
The effect of laser shock processing on the microstructures and properties of 2060 Al-Li alloys
Surf Coat Technol, 434 (2022), Article 128208
View PDF View article View in Scopus ↗ Google Scholar ↗

[27] Z.B. Shen, H.X. Liu, X. Wang, *et al.*
Improving the forming capability of laser dynamic forming by using rubber as a forming medium
Appl Surf Sci, 369 (2016), pp. 288-298
View PDF View article View in Scopus ↗ Google Scholar ↗

[28] C. Wang, X.G. Wang, Y.J. Xu, *et al.*
Numerical modeling of the confined laser shock peening of the OFHC copper
Int J Mech Sci, 108-109 (2016), pp. 104-114

View PDF View article Google Scholar

[29] C.S. Montross, T. Wei, L. Ye, *et al.*
Laser shock processing and its effects on microstructure and properties of metal alloys: a review
Int J Fatigue, 24 (10) (2002), pp. 1021-1036
View PDF View article View in Scopus Google Scholar

[30] S. Caruso, M. Sanguedolce, G. Serratore, *et al.*
Lagrangian finite element model formulation and experimental validation of the laser impact weld process for Ti/brass joining
J Manuf Mater Process, 8 (2024), Article 141
Crossref View in Scopus Google Scholar

[31] B. Jiang, Z.S. Guo
Determination traction-separation parameters and its sensitivity analysis of bilinear cohesive zone model through experimental and numerical peeling analysis of electrodes
Int J Adhes Adhes, 133 (2024), Article 103760
View PDF View article View in Scopus Google Scholar

[32] A. Jaillon, J. Jumel, F. Lachaud, *et al.*
Mode I cohesive zone model parameters identification and comparison of measurement techniques based on uncertainty estimation
Int J Solids Struct, 191-192 (2020), pp. 577-587
View PDF View article View in Scopus Google Scholar

[33] L.A. de Oliveira, M.V. Donadon
Delamination analysis using cohesive zone model: a discussion on traction-separation law and mixed-mode criteria
Eng Fract Mech, 228 (2020), Article 106922
View PDF View article View in Scopus Google Scholar

[34] B.C. Liu, A.N. Palazotto, A. Nassiri, *et al.*
Experimental and numerical investigation of interfacial microstructure in fully age-hardened 15-5 PH stainless steel during impact welding
J Mater Sci, 54 (2019), pp. 9824-9842
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[35] R.C. George, H.H. Arnold
Flow configurations in colliding plates: explosive bonding
J Appl Phys, 34 (1963), pp. 928-939
Google Scholar ↗

[36] A. Nassiri, B. Kinsey, G. Chini
Shear instability of plastically-deforming metals in high-velocity impact welding
J Mech Phys Solids, 95 (2016), pp. 351-373
View PDF View article View in Scopus ↗ Google Scholar ↗

[37] B.A. Greenberg, M.A. Ivanov, A.V. Inozemtsev, *et al.*
Comparative characterisation of interfaces for two- and multi-layered Cu-Ta explosively welded composites
Compos Interfaces, 27 (2020), pp. 705-715
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[38] R. Shotri, K. Faes, A. De
Magnetic pulse welding of copper to steel tubes-experimental investigation and process modelling
J Manuf Process, 58 (2020), pp. 249-258
View PDF View article View in Scopus ↗ Google Scholar ↗

[39] P. Shewmon
Diffusion equations
Diffusion in solids, Springer, Cham (2016), pp. 14-22

Google Scholar ↗

[40] A.X. Mao, J.P. Zhang, S.C. Yao, *et al.*
The diffusion behaviors at the Cu-Al solid-liquid interface: a molecular dynamics study
Results Phys, 16 (2020), Article 102998
View PDF View article View in Scopus ↗ Google Scholar ↗

[41] T. Shanmugasundaram, M. Heilmaier, B.S. Murty, *et al.*
On the Hall-Petch relationship in a nanostructured Al-Cu alloy
Mater Sci Eng A, 527 (2010), pp. 7821-7825
View PDF View article View in Scopus ↗ Google Scholar ↗

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Michel Michaelides, ..., Anthony T Moore

Taylor and Hoyt's Pediatric Ophthalmology and Strabismus • 2017 • Pages 462-486.e2

View PDF

Chapter Full text access

12 - Outcomes assessment

Timothy P. Melchert

Foundations of Health Service Psychology • 2020 • Pages 221-236

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